

```

if [[ -e "$BACKUP_DIR"/doc_root_backup-$(date +%d-%m-%y).tar.bz2 ]]; then
    backupDatabase
else
    echo "Backup
failed for document root on your server. Please check
manually" | mail -s "Backup failed - Document Root" user@
domain.com
fi
}

function backupDatabase() {
    `which mysqldump` -u "$DB_USER" -p"$DB_PASSWORD"
"$DB_NAME" > "$BACKUP_DIR"/"$DB_NAME"-"$(date +%d-%m-%y).sql |
tee "$BACKUP_DIR"/backup.log
    tar -cjf "$BACKUP_DIR"/"$DB_NAME"-"$(date +%d-%m-%y).tar.bz2
"$BACKUP_DIR"/"$DB_NAME"-"$(date +%d-%m-%y).sql
--remove-files 2> $BACKUP_DIR/tar.log
    if [[ -e "$BACKUP_DIR"/"$DB_NAME"-"$(date +%d-%m-%y).tar.bz2
]]; then
        createTar
    else
        echo "Backup
failed for database on your server. Please check manually" |
mail -s "Backup failed - Database" user@domain.com
    fi
}

function createTar() {
    cd "$BACKUP_DIR"
    `which tar` -cf backup-"$(date +%d-%m-%y).tar doc_
root_backup-$(date +%d-%m-%y).tar.bz2 "$DB_NAME"-"$(date
+%d-%m-%y).tar.bz2 --remove-files 2> final.log
                                deleteOldBackup
}

function deleteOldBackup() {
    `which find` "$BACKUP_DIR"/"$DB_NAME"* -mtime +$DAYS_
TO_KEEP_BACKUP" -exec rm {} \; >> "$BACKUP_DIR"/delete.log
}

main
# End

```

Editable options

Given below is the basic information that a user can input in the script.

1. **BACKUP_DIR**: This is the directory in which backups (database and document root) will be placed.
2. **DB_USER**: This is the username of the database that you want to take the backup of.
3. **DB_PASSWORD**: This is the associated password for the username mentioned in the second point.
4. **DB_NAME**: This is what the database that you want to

take a backup of is called.

5. **DOC_ROOT**: This is the path to the document root where your website's files and directories reside. For example, if your website's content is stored at `/usr/share/nginx/html/my_website`, then this path has to be defined in the **DOC_ROOT** variable.
6. **DAYS_TO_KEEP_BACKUP**: This is self-explanatory. The backup will be kept on the server for the number of days mentioned here. By default, I have set it to five days. Any backup file that is older than that will be removed automatically.

An explanation

This script is modular in nature, which makes it easy to maintain and make changes to add or remove various functionality. Given below is the basic overview:

1.

```

function main() {
    init
    chkBackupDir
}

```

A function named *main* is defined, which contains other functions called *init* and *chkBackupDir*. As you will see below, these two functions (defined inside the main function) have different functionalities and the *main* function just calls them.

2.

```

function init() {
    BACKUP_DIR="/path/to/backup/directory"
    DB_USER="db_user"
    DB_PASSWORD="db_password"
    DB_NAME="db_name"
    DOC_ROOT="/path/to/document/root"
    DAYS_TO_KEEP_BACKUP="5"
}

```

The *init* function is defined, which contains all the variables that need to be initialised when the script runs. Each variable is explained above.

3.

```

function chkBackupDir() {
    if [[ ! -d "$BACKUP_DIR" ]]; then
        mkdir "$BACKUP_DIR"
    fi
    backupDocRoot
}

```

The *chkBackupDir* function is defined, which will check if the backup directory (mentioned by the user in the variable) is present or not. If not, it will be created and